CPA1 - MAN AND MAN ENTERPRISE IMPROVED COOKING STOVES PROGRAMME IN GHANA (BRONG-AHAFO REGION)

Document Prepared by AERA Group

On behalf of Man and Man Enterprise

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The purpose of the proposed activity "CPA1 - Man and Man Enterprise Improved Cooking Stoves Programme in Ghana (Brong-Ahafo region)", hereafter referred to as 'CPA1', is to combat climate change and contribute to sustainable development of Ghana.

In the baseline scenario, households continue to using non-renewable biomass in traditional cooking stoves. An ICS combusts wood fuel more efficiently, i.e. requires less charcoal than a traditional stove. This reduces CO_2 emissions.

CPA1 thus aims to reduce non-renewable wood fuel consumption and greenhouse gas (GHG) emissions of households (hereafter also "end-users") in the urban and peri-urban Brong-Ahafo region of Ghana by selling 203,460 affordable Improved Cooking Stoves (ICSs) in replacement of traditional charcoal cooking stoves, so-called coal pots, resulting in 174,774 tCO_2 emission reductions per year on average or 1,223,420 tCO_2 over the 7-year crediting period.

The CPA implementer is Man and Man Enterprise, a Ghana-based ICS producer and seller. The ICS distributed under CPA1 is the so-called Jiko-type ICS. It is also known as the "Holy cook", which is the brand name of the Jiko-type ICS produced by Man and Man Enterprise. Compared to coal pots, which have an efficiency of 15%-18%, Holy Cooks have an efficiency of about 30% while providing the same service. This allows better heat retaining, i.e. quicker heating-up and longer cooking times with less wood fuel (and combustion fumes). ICSs work on a simple but mature technology for efficient energy conversion of biomass to heat. Furthermore, the stoves distributed have a lifetime of 5[1] years as per registered CPA documentation.

Initially undertaken as CDM PoA 10430 "Man and Man Enterprise Improved Cooking Stoves CDM Programme in Ghana supported by Republic of Korea", the small-scale project type applicable to the CPA is Type II, i.e. an energy efficiency improvement project activity "that reduce energy consumption, on the supply and/or demand side, with a maximum energy saving of 60 GWh per year (or an appropriate equivalent) in any year of the crediting period."

Yet, the CPA qualifies as a microscale project type II which units aims to achieve energy savings at a scale of no more than 600 MWh per year, which is equivalent to 1,800 MWh_{th} of annual energy savings per appliance. During this monitoring period from January 1st 2021 to May 31st 2022, the project resulted in 402,057 tCO_2e emission reductions.

Audit Type	Period	Program	VVB Name	Number of years

[1] Stoves with 5 years or more from the sale dates, will be removed from the database, and no emissions reductions, will be requested from them even if they are found working.

Validation & Verification	01-January-2021 to 31-May-2022	**VCS**	**Carbon Check (India) Private Ltd.**	17 months (1 year, 5 months) : 402,057 tCO_2e
Total[2]	01-January-2021 to 31-May-2022	-	-	17 months (1 year, 5 months) : 402,057 tCO_2e

1.2 Sectoral Scope and Project Type

Sectoral Scope : 03 Energy demand
Project Type : Type II Energy Efficiency Improvement Project
Methodology used: AMS II.G Ver.08.0 Energy efficiency measures in thermal applications of non-renewable biomass.
Grouped Project: This is a single project activity.

AFOLU project: This is not an AFOLU project

1.3 Project Eligibility

As per para. 2.1.1 of the VCS Standard, V.4.4, the project is eligible under the scope of the VCS Program because the VCS Scope includes project activities involving the following elements:

Elements	**Project applicability**
The six Kyoto Protocol greenhouse gases	The project activity and AMS II.G V.08.0 aims at CO_2 reduction of non-renewable biomass.
Ozone-depleting substances.	n/a
Project activities supported by a methodology approved under the VCS Program through the methodology approval process.	n/a
Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval.	CDM methodology AMS II.G, V.08 is applied. The CDM is a VCS approved GHG program.[3] Any methodology developed under the United Nations Clean Development Mechanism can be used for projects and programs registering with VCS.[4]
Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements.	n/a

[2] Verra programs only are represented in this table, therefor the project's previous validation and credits issued under CDM are not visible. For more information on the project's history under this program, see section 1.15

[3] https://verra.org/project/vcs-program/projects-and-jnr-programs/develop-a-project/
[4] https://verra.org/methodologies/.

Also, as per APPENDIX 2 ELIGIBILITY CONDITIONS FOR PROJECTS AND CPAS SEEKING VCS REGISTRATION of VCS Standard 4.4, following eligibility criteria also applies for CDM CPAs seeking registration under the VCS Program when they don't include afforestation and/or reforestation activities:

Eligibility criteria	Compliance with the eligibility criteria
The CPA must be part of a Program of Activities (PoA) with an original program crediting period start date on or after 1 January 2016;	Applicable. The CPA requesting for gap validation is included in the PoA 10430: Man and Man Enterprise Improved Cooking Stoves CDM Programme in Ghana supported by Republic of Korea which has a registration date on 31 May 2018 and a first crediting period spanning from 31 May 2018 to 30 May 2025. Furthermore, the CPA has an inclusion date on 30 November 2018 and a first crediting period spanning from 30 November 2018 to 29 November 2025. Hence the CPA complies with this criteria
Where the CPA is part of a Program of Activities (PoA) with an original program crediting period start date from 1 January 2013 to 31 December 2015 and where the CPA has an original crediting period start date from 1 January 2013 to 31 December 2015, the CPA must have issued credits during the period 1 January 2016 to 5 March 2021, or must have a status of "issuance requested" by 5 March 2021; or	Not applicable as the PoA and the CPAs have both crediting period start after 31 December 2015
Where the CPA is part of a PoA with an original program crediting period start date from 1 January 2013 to 31 December 2015 and where the CPA has an original crediting period start date on or after 1 January 2016, no prior credit issuance is required.	Not applicable as the PoA and the CPAs have both crediting period start after 31 December 2015
For a CPA that does not include afforestation and/or reforestation activities, only emission reductions with vintages beginning on or after 1 January 2016 are eligible for VCU issuance.	Applicable. The vintage being request are the vintage 2021 and 2022, hence after 2016 batches.

Furthermore, it cannot reasonably be assumed that the programme generates GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction. All users involved in the programme activity are fully independent from the project proponent.

1.4 Project Design

The project occurs in the limits of Brong-Ahafo region now divided in Bono and Bono East regions. Therefore, the project occurs in multiple locations of these regions, and is developed as a single project activity. All the ICS are distributed in the limits of Brong-Ahafo region (see map below).

1.5 Project Proponent

Organization name	Man and Man Enterprise
Contact person	Mr. Agyei Michael Yaw
Title	CEO
Address	Kumasi Santasi New site-Kumasi Kumasi, Ashanti region Ks 4588
Telephone	+233 (0) 243 47 36 42
Email	manandman.ent@gmail.com

1.6 Other Entities Involved in the Project

Organization name	AERA Group
Role in the project	Carbon consultant and trader
Contact person	Alexandre Dunod
Title	Chief Operating Officer
Address	28 Cours Albert 1er, 75008 Paris, France
Telephone	+33 1 42 18 02 02
Email	a.dunod@aera-group.fr

1.7 Ownership

As per VCS Program Definitions version 4.1, the project ownership is the legal right to control and operate the project activities. As project implementer, Man & Man have the full ownership of the project. Furthermore, as per VCS Standard V4.2 para 3.6.1, the project description shall be accompanied by one or more of the following types of evidence establishing project ownership accorded to the project proponent(s), or program ownership accorded to the jurisdictional proponent(s), as the case may be (see the VCS Program document Program Definitions for definitions of project ownership and program ownership). The type of evidence to prove project ownership is point 5 of para 3.1.6 :"An enforceable and irrevocable agreement with the holder of the statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals which vests project ownership in the project proponent".

All the end-users (technology owners) sign a stove sales agreement, which contains user information, rights and commitments including relinquishing ERs ownership to Man and Man.

Also, Man and Man has entered in agreement with AERA Group, in which they cede rights to sell ERs generated by their programme. End-users' agreements and agreement extract between Man and Man and AERA are available and will be provided to DOE to prove project ownership.

1.8 Project Start Date

Start date of the project is 20/10/2017, as the sales date of the first ICS.

1.9 Project Crediting Period

The CPA has an expected lifetime of 21 years, as the crediting period is 7 years and is renewable twice. The current crediting period started on the 20/10/2017 and ends on 19/10/2024.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	X
Large project	

Year	Estimated GHG emission reductions or removals (tCO_2e)
2017	0[5]
2018	1,961
2019	70,393
2020	171,943
2021	251,264
2022	251,264
2023	251,264
2024	225,330

[5] These values slightly differ from the ex-ante ERs visible in the CDM registered CPA-DD, since the start date had to be adapted in regard to the VCS gap validation. Also, the volumes of ICS sold were realigned with the database.

Total estimated ERs	1,223,420
Total number of crediting years	7
Average annual ERs	174,774

1.11 Description of the Project Activity

n/a. Please see registered CDM-CPA (as per para. 3.22.6 of VCS Project Standard V.4.4).

Link to the CPA:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ/view

1.12 Project Location

CPA1 is implemented in the Brong-Ahafo region of Ghana consisting of 21 districts.


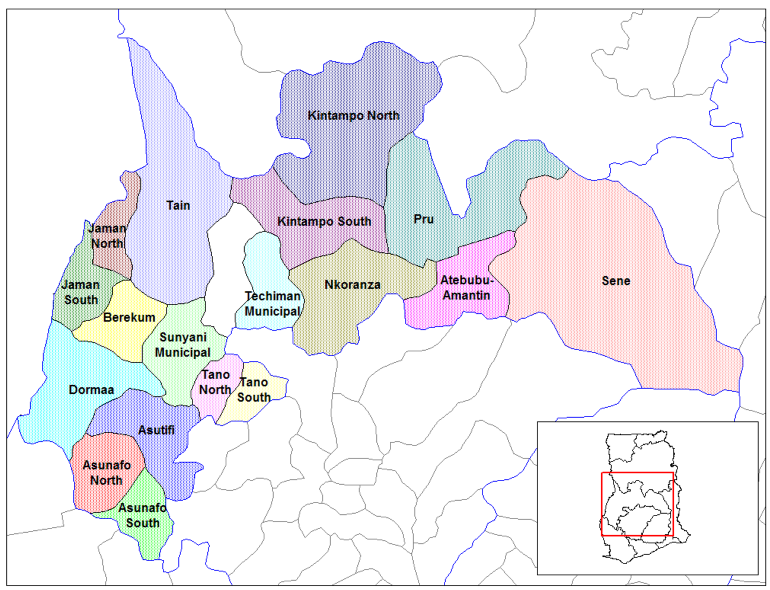

Figure 1 : Maps of Brong-Ahafo region in Ghana (left) and its districts (right)

The GPS coordinates of the capital Sunyani in the Sunyani Municipal district are 7°44' 60" N - 1°29'60" W (07.7500000°, -001.5000000°).

1.13 Conditions Prior to Project Initiation

The conditions prior to the project initiation are the same as the baseline scenario.

The project activity involves dissemination of higher efficiency biomass fired cook stoves use as household appliances for cooking purpose. This will contribute to the reduction of non-renewable biomass consumption which would have been otherwise consumed by traditional three stone open firing, less efficient cook stoves.

Please refer to section B.3 pf CDM-CPA-DD of the UNFCCC registered CDM PoA Ref No. 10430.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project activity does not have any regulatory requirements within the republic of Ghana, and all activities associated with the project will be undertaken in compliance with local business and social regulations and requirements. A review of relevant laws of Ghana did not highlight any specific concerns or requirements around cookstoves or any aspects related to this project activity.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project activity has already been registered under Clean Development Mechanism (CDM) of UNFCCC, under Kyoto Protocol. This project is a Component Project Activity (CPA) under a registered Programme of Activities (CDM PoA). The details are as follows:

CDM-PoA Registration number : 10430

Title: Man and Man Enterprise Improved Cooking Stoves CDM Programme in Ghana supported by Republic of Korea

Weblink:
https://cdm.unfccc.int/ProgrammeOfActivities/poa_db/KQXLWC1G6IEY8OHVDFU9S27T5ZNMRP/view

CPA Reference number : 10430-P1-0001-CP1

Title: Improved Cooking Stoves Programme in Burundi supported by Republic of Korea CPA1

Weblink:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ/view

Previous issuance requests :

- Monitoring report: 30/11/2018 – 13/12/2018 | CERs requested : 611
- Monitoring report: 14/12/2018 – 13/12/2019 | CERs requested : 55,895
- Monitoring report: 14/12/2019 – 31/12/2020 | CERs requested : 160,401

The project will never verify nor claim ER from both VCS and CDM standards on a same period.

Applicability criteria for CDM projects :

• This CPA1 registered under the CDM have the same project activity and project proponent than the CPA2 also registered under the CDM, however they are occurring more than 10 km from each other, as the only common barrier between the two regions is covered by the Digya national Park, and the zone does not have any habitat (as shown in the KML file provided to VVB)

• This CPA1 only seeks to be registered under the VCS

• All requested sections of the VCS-PDD are completed in this Joint PD-MR document

• This CPA1 is not subdivided into smaller projects or combine subdivided CPAs into one VCS project, as the boundary stays the same as under CDM.

1.15.2 Projects Rejected by Other GHG Programs

The project has never been rejected by another form of GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Ghana is non annex 1 country and there is no compliance with an emission trading program or to meet binding limits on GHG emissions for this project activity. The project activity has been registered as a CPA under a CDM PoA (Ref. 10430) and GHG emission reductions and removals generated by the project will not be used for any compliance under any programs or mechanism, except CDM. The project activity under VCS shall not claim emission reductions for the same period that has been/will be covered under CDM as a part of the registered PoA. Thus, there will not be any possibility of double accounting of emission reductions.

1.16.2 Other Forms of Environmental Credit

The project has neither sought not received another form of GHG-related environmental credits.

1.16.3 Supply Chain (Scope 3) Emissions

Project proponent of this project, Man and Man Enterprise, is also the manufacturer and distributor of the project devices. They are also responsible for the collection of raw material like scrap metal and clay. In this regard, the only stakeholder in the supply chain of this project concerned by the carbon financing acknowledgement are the end-users who already make such pledge in the sales agreement at the time of the stove purchase. The sales agreements contain this statement since the start of the project activity and avoid potential double counting of the emission reductions.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

The project activity, while providing efficient cookstoves to households in Ghana, will have positive impacts on sustainable development: fuel consumption is reduced for cooking, and this reduces the amount of non-renewable biomass used in the region (SDG 15.2). According to the Global Forest Watch, Ghana has an annual deforestation and forest degradation rate of 2%, equivalent to 135,000 hectares loss of forest cover. Another beneficial SDG impact is clean energy access at an affordable cost (SDG 7.1), as the project which aims the distribution of 203,460 ICS actively participates in that regard. Also, the project will result in the reduction of Household Air Pollution (HAP) in rural household (SDG 3.9.1): indeed, the World Health Organisation (WHO) confirms that a more efficient combustion has a positive effect on health[6], since indoor pollutants – specially VOCs – are reduced. In Ghana's updated National Determined Contribution, the country confirmed its will to expand the adoption of clean cooking solutions as well as the overall improvement of air quality by 2030.

On the social aspect, the project will also reduce the proportion of time rural women spend on unpaid domestic work related to collection of fuel, and cooking meals on inefficient traditional cookstoves (SDG 5.4.1). The gained time can then be applied elsewhere in economic or social pursuits.

The project also presents economic benefits, in a region where expenses on cooking fuel are significant in the household resources. The reduction of fuel needed in an ICS will result in lower expenses in that regard (SDG 1.1). The production of cookstoves will also create employment in the region (SDG 8.5).

Finally, the GHG emissions reduction is estimated at 1,198,634 tCO_2e during the first crediting period. These avoided emissions actively participate in the urgent action needed to mitigate climate change (SDG 13).

1.17.2 Sustainable Development Contributions Activity Monitoring

Among those multiple Sustainable Development Goals positive impacts, the project will monitor at each monitoring period a total of 5 of them. The core activity of this project which is the dissemination of improved cooking stoves, creates jobs for the production, distribution and monitoring of the units. It improves the health of the users since the use of the improved stove reduces air pollution. Also, the reduction of fuel needed for cooking result in saving for end-users, as well as CO_2 emission reductions.

The project activity contributes to achieving nationally stated sustainable development priorities : among them, the creation of decent jobs, avoidance of premature deaths (air quality) and the generation of absolute greenhouse gas emission reduction by 2030[7].

[6] Household air pollution and health. Available at : https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health

[7] MESTI. (2021). Ghana: Updated Nationally Determined Contribution under the Paris Agreement (2020 – 2030) Environmental Protection Agency, Ministry of Environment, Science, Technology and Innovation, Accra.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	1.1.	Household confirmation of money savings due to reduced spendings	Use of stoves results in money savings for end-users (less biomass fuel needed)	96.0% of the monitoring survey respondents have confirmed money savings on fuel purchase since they use the Man&Man stove	During previous monitoring periods, end-users were not asked regarding money savings in the monitoring surveys.
2)	3.9	Household perception of health benefits (reduction of inhaled smokes)	Distribution of stoves results in less harmful smokes (higher efficiency + less biomass fuel needed)	96.0% of the monitoring survey respondents have confirmed they felt less exposed to smoke since they use the Man&Man stove.	During previous monitoring periods, end-users were not asked regarding smoke exposure in the monitoring surveys.
3)	7.1	7.1.1 Proportion of population with primary reliance on clean fuels and technology	Distribution of stoves results in more efficient and cleaner fuels use for end-users, compared to the baseline scenario	195,980 project devices operating during this monitoring period. (Calculated from the number of stoves distributed under this project activity multiplied by the percentage of operating devices during for this monitoring period)	203,460 project devices distributed since the start of the project activity.
4)	8.5	8.5.2 Unemployment rate, by sex, age, and persons with disabilities	Job creation for the production, distribution and monitoring of the stoves under this project activity	10 employees currently working at Man&Man Enterprise. No new job created during this monitoring period.	10 jobs created since the start of the project activity.
5)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Use of stoves by end-users results in avoided greenhouse gas emissions	402,061 tCO_2 avoided during this monitoring period	618,964 tCO_2 avoided since the start of the project activity MP1 : 611 MP2 : 55,895 MP3 : 160,401 MP4 : 402,057

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable to this project activity.

The project activity has been designed to include only new improved cookstoves. No used improved cook stoves will be transferred or deployed from outside the geographical project boundary to the project activity. Also, the project activity applies CDM Methodology: AMS-II.G. "Energy Efficiency Measures in Thermal Applications of Non-Renewable Biomass", Version 08.0. According to the para 32 of the methodology, the default net to gross adjustment factor of 0.95 has been applied to account for leakage and therefore no surveys or no separate leakage management are required. This information has been further referred under the registered CPA-DD, under the section B.4.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not applicable.

2 SAFEGUARDS

2.1 No Net Harm

n/a. Please see registered CDM-CPA (as per para. 3.22.6 of VCS Project Standard V.4.4)

Link to the CPA:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ/view.

2.2 Local Stakeholder Consultation

n/a. Please see registered CDM-CPA (as per para. 3.22.6 of VCS Project Standard V.4.4)

Link to the CPA:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ/view.

Ongoing communication with local stakeholders is assured to allow stakeholders to raise concerns about potential negative impacts of the project. The following mechanisms are established :

- Contact details of Man&Man enterprise (including address, telephone number and e-mail address) are described in the sales agreements.

- Grievance logbook located in the offices of Man&Man In Kumasi
- Direct feedback from end-users in the Brong-Ahafo region to the monitoring and replacement staff from Man&Man, who do regular site visits.

During the monitoring period from 01/01/2021 to 31/05/2022, Man&Man enterprise has not received any e-mail from stakeholders concerning this project.

2.3 Environmental Impact

n/a. Please see registered CDM-CPA (as per para. 3.20.5 of VCS Project Standard V.4.2) section D

Link to the CPA:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ

2.4 Public Comments

For the gap validation/verification under VERRA, this project was open for public comment from 07/09/2022 to 07/10/2022. No comments have been received.

2.5 AFOLU-Specific Safeguards

The project is a non-AFOLU project activity, hence not applicable.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

n/a. Please see registered CDM-CPA (as per para. 3.22.6 of VCS Project Standard V.4.4) section B

Link to the CPA:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ/view

3.2 Applicability of Methodology

n/a. Please see registered CDM-CPA (as per para. 3.22.6 of VCS Project Standard V.4.4) section B

Link to the CPA:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ/view

3.3 Project Boundary

n/a. Please see registered CDM-CPA (as per para. 3.22.6 of VCS Project Standard V.4.4) section B

Link to the CPA:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ/view

3.4 Baseline Scenario

n/a. Please see registered CDM-CPA (as per para. 3.22.6 of VCS Project Standard V.4.4) section B

Link to the CPA:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ/view

3.5 Additionality

n/a. Please see registered CDM-CPA (as per para. 3.22.6 of VCS Project Standard V.4.4) section B

Link to the CPA:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ/view

3.6 Methodology Deviations

No deviations have been made therefore this section is not applicable.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

During this monitoring period, the project activity operated well and continuously, whether on the manufacturing or the distribution side of the project devices. The total number of stoves sold during the monitoring period i.e. from 01/01/2021 to 31/05/2022 amounts to 37,039 stoves.

Since the project start, 203,460 stoves have been distributed (see table 2 and 3 below). Among those 203,460 stoves, 195,980[8] project devices were found operating during this monitoring period. These belong to three batches: 2018 (spanning from 04/06/2018 to 03/06/2019), 2019 (spanning from 04/06/2019 to 03/06/2020) and 2020 (spanning from 04/06/2020 to 03/06/2021). The last stoves distributed were sold on the 07/05/2021, and they belong to the 2020 batch. The distribution has stopped since that date.

Table 2 : Distribution per batch

Batch	Number of stoves distributed
2018 (04/06/2018-03/06/2019)	35,142
2019 (04/06/2019-03/06/2020)	69,870
2020 (04/06/2020-03/06/2021)	98,448[9]
Total	**203,460**

Table 3: distribution per year

Year	Number of stoves distributed
2018	20,993
2019	41,588
2020	103,840
2021	37,039
Total	**203,460**

During this monitoring period, 3 comments were written in the grievance logbook, regarding the damages on their stoves and design of ceramic liner. The stoves of the 3 end-users have been replaced as part of a monitoring visit.

For replacement, the stove ID is removed from the damaged stove, and fixed on the new stove. The old stove is then transported to Man&Man offices, where the stoves are dissembled, and if the metal part is found in good shape, it is reused to make new stoves. For replacements, the sale date does not change, conservatively the first distribution date is kept for ERs calculations.

[8] Taking in consideration a mean of operation of 96.32% for all stoves.

[9] During the last monitoring period under the CDM, a total of 61,407 stoves were distributed by the end of 2020 for Batch 2020. An additional 37,039 stoves were distributed in 2021, bringing the total for Batch 2020 to 98,446 stoves. The discrepancy of two stoves between the numbers 98,446 and 98,448 arises because these two stoves (IDs 2M.27228 and 3M.11195), originally distributed under Batch 2019, were lost by end-users and subsequently replaced. These replacements have been recorded under Batch 2020. For the 2019 Batch, the verified number of Improved Cook Stoves (ICS) during the 3rd CDM verification was 69,872, compared to 69,870 reported in this monitoring period. For conservative reasons, these two stoves are not included in the emission reduction (ER) calculation. Please see cell M61 of the VCU calculation tab in the ER sheet for more details.

However, when a stove is lost or stolen, in opposite of damaged stoves, the stove is replaced and the date of replacement is used as distribution date.

5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

n/a. Please see registered CDM-CPA (as per para. 3.22.6 of VCS Project Standard V.4.4) section B.4

Link to the CPA:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ/view

5.2 Project Emissions

As per AMS II.G v08.0 methodology, Emissions reductions are calculated directly. Therefore, this section is not applicable.

5.3 Leakage

n/a. Please see registered CDM-CPA (as per para. 3.22.6 of VCS Project Standard V.4.4) section B.4

Link to the CPA:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ/view.

5.4 Estimated Net GHG Emission Reductions and Removals

n/a. Please see registered CDM-CPA (as per para. 3.22.6 of VCS Project Standard V.4.4) section B.4

Link to the CPA:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ/view

6 MONITORING

6.1 Data and Parameters Available at Validation

n/a. Please see registered CDM-CPA (as per para. 3.22.6 of VCS Project Standard V.4.4) section B.4.2

Link to the CPA:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ/view

6.2 Data and Parameters Monitored

n/a. Please see registered CDM-CPA (as per para. 3.22.6 of VCS Project Standard V.4.4) section B.5

Link to the CPA:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ/view

6.3 Monitoring Plan

n/a. Please see registered CDM-CPA (as per para. 3.22.6 of VCS Project Standard V.4.4) section B.5

Link to the CPA:
https://cdm.unfccc.int/ProgrammeOfActivities/cpa_db/R71MGTX26FAZ93OE4BWKI80VSUPHCJ/view

The equation (6) of the CPA-DD section B.5 was used for the calculation of the minimum sample size for monitoring surveys on a 95/10 level of precision/confidence. Then, a 20% factor (oversampling) was applied to the values. For each batch, it was established a minimum of 90 stoves needed to be monitored, 270 in total. In the end, the actual sample size amounts to 272, with at least 90 stoves per batch. The stoves were selected through a program generating random numbers and owners contacted to prepare and confirm their availability before the monitoring visit. The monitoring surveys were completed from May 3rd 2022 until August 15th 2022. Details of the sampling calculation is displayed in the ER sheet.

The equation (7) of the CPA-DD section B.5 was used for the calculation of the minimum sample size for water boiling tests on a 90/10 level of precision/confidence (annual). Then, a 20% factor (oversampling) was applied to the values. For each batch, it was established a minimum of 2 stoves needed to be tested, 6 in total. In the end, the actual sample size amounts to 9, corresponding to 3 per batch. The stoves were selected through a program generating random numbers and owners contacted to prepare and confirm their agreement beforehand. A stove was lent to each owner in temporary replacement of the stove being tested in lab. When lab testing is over, the lent stoves is taken back to the Man&Man offices and original stove given back to the owner. Details of the sampling calculation is displayed in the ER sheet.

Man & Man Enterprise has frequent site visit and data check as part of its internal auditing. Project proponent trained monitoring personnel on monitoring procedures, including provisions for maximizing response rates, documenting out-of-population cases, refusals and other sources of non-response. The monitoring survey included several questions to support the information on the key monitoring parameters. These included visual inspections to confirm stove use and presence of baseline stoves, comments by surveyors, check of randomly selected households against actual household information, and refusal tracking. These strategies were aimed at minimizing surveyor or non-response biases.

During the onsite visit by the VVB, it was observed that 2 stoves had stickers and ID numbers pasted from a different project activity implementer(SUDRA) operating in the project area. Project developers, and beneficiaries clarified that those stoves were well manufactured and distributed by Man&Man, and that the stickers were illegitimately applied on the stoves by SUDRA, several measures have been taken in order to tackle the double-counting risk:

- For 1 month, a monitoring team in charge of replacements have been checking for other stoves in the same situation. This team comprised 11 Man&Man workers who operated in 20+ districts/towns, in Q4 of 2022. While thousands of stoves were checked and 2,418[10] were replaced, only 14 stoves with double IDs were identified during this campaign.
- The 14 involved stoves with double IDs have been replaced following this campaign. A second onsite audit took place to confirm that the problem was solved.
- A communication with the Man&Man and SUDRA was initiated on VVB request. SUDRA recognized the error from one of their workers, confirmed that those stoves are not registered in their database, and that no ERs are never requested for those, hence excluding the risk of double counting. Finally, they signed and shared a letter confirming these claims. Such letter has been shared to VVB.

Following this:

Man&Man has taken the following measure to ensure that this issue is not found again:

a. An extended awareness campaign has been done to inform the households that their stoves should only be maintained by Man&Man team, and that no other project developer should be authorized to work on their stoves. They can get in touch physically or by phone with Man&Man team in case any doubts.

b. Increase the frequency and target of follow-up visits to detect any issues regarding double identification.

c. Discussion with project developers in project areas to ensure that this issue does not repeat.

Details of the sampling size calculation per batch and for different parameters can be found in the tables below (taken from ex-post ER sheet) :

2018 Batch

Parameters	Description	level of confidence / precision	t (α/2) Student t Critical	Size of the population of stoves considered for the monitoring session	Expected proportion (p)			Sample size before oversampling	Sample size after oversampling
pop stoves,y	Proportion of distributed ICS still operating or replaced by an equivalent in service appliance under similar conditions under the project activity in year y	90 / 10	1,645						
		95 / 10	1,96	35 142	97,22%			11,00	14,00
μ,y (pre-project devices use)	Proportion of residual use of woody biomass from non-renewable sources under the project activity in year y	90 / 10	1,645						
		95 / 10	1,96	35 142	83,69%			75,00	90,00
	Description	level of confidence / precision	t (α/2) Student t Critical	Size of the population of stoves considered for the monitoring session	Standard Deviation and Mean	Equation for V	V		
$\eta_{new,y}$	Efficiency of the device being deployed as part of the project activity in year y	90 / 10	1,645	35 142	0,015	$\left(\frac{standard\ deviation}{mean}\right)^2$	0,002	1,00	2,00
		95 / 10	1,96		0,32				

[10] These 2,418 stoves have been replaced after the actual monitoring period, hence not reported in the section 4.1. The stoves replaced, are the ones found in bad shape, following the campaign conducted after the double ID issue. The list of all the stoves replaced has already been shared with VVB, and this will also be reported in upcoming monitoring period. All the stoves replaced have kept their original sale date as per replacement guidelines explained in section 4.1

2019 Batch

Parameters	Description	level of confidence / precision	t (α/2) Student t Critical	Size of the population of stoves considered for the monitoring session	Expected proportion (p)			Sample size before oversampling	Sample size after oversampling
pop stoves,y	Proportion of distributed ICS still operating or replaced by an equivalent in service appliance under similar conditions under the project activity in year y	90 / 10	1,645						
		95 / 10	1,96	69 870	99,01%			4,00	5,00
μ,y (pre-project devices use)	Proportion of residual use of woody biomass from non-renewable sources under the project activity in year y	90 / 10	1,645						
		95 / 10	1,96	69 870	83,69%			75,00	90,00
	Description	**level of confidence / precision**	**t (α/2) Student t Critical**	**Size of the population of stoves considered for the monitoring session**	**Standard Deviation and Mean**	**Equation for V**	**V**		
$\eta_{new,y}$	Efficiency of the device being deployed as part of the project activity in year y	90 / 10	1,645	69 870	0,015	$\left(\frac{standard\ deviation}{mean}\right)^2$	0,002	1,00	2,00
		95 / 10	1,96		0,32				

2020 Batch

Parameters	Description	level of confidence / precision	t (α/2) Student t Critical	Size of the population of stoves considered for the monitoring session	Expected proportion (p)			Sample size before oversampling	Sample size after oversampling
pop stoves,y	Proportion of distributed ICS still operating or replaced by an equivalent in service appliance under similar conditions under the project activity in year y	90 / 10	1,645						
		95 / 10	1,96	98 448	99,01%			4,00	5,00
μ,y (pre-project devices use)	Proportion of residual use of woody biomass from non-renewable sources under the project activity in year y	90 / 10	1,645						
		95 / 10	1,96	98 448	83,69%			75,00	90,00
	Description	**level of confidence / precision**	**t (α/2) Student t Critical**	**Size of the population of stoves considered for the monitoring session**	**Standard Deviation and Mean**	**Equation for V**	**V**		
$\eta_{new,y}$	Efficiency of the device being deployed as part of the project activity in year y	90 / 10	1,645	98 448	0,015	$\left(\frac{standard\ deviation}{mean}\right)^2$	0,002	1,00	2,00
		95 / 10	1,96		0,32				

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	-
Description	Number of project devices of type i and batch j operating during year y
Value applied:	195,980
Comments	Calculated by multiplying parameter N Number of project devices distributed by monitored percentage of operational stoves. Parameter from CPA monitoring database. If baseline stove is not included under baseline defined under the specific CPA, the new device is counted as not operating, i.e. no emission reductions are claimed.

% of operational stoves precision levels, per batch :

2018 :

Ope_stove average	**91.11%**
Sample size	**90**
Total population size	**35,142**
Required precision	**95%**
z-value at 90% confidence	**1.96**
Confidence interval (+/-)	**6%**
Lower bound of the interval	**85%**
Higher bound of the interval	**97%**
Maximum error (precision)	**10%**
Sample monitoring precision	**6%**
Conclusion	**Precision OK**

2019 :

Ope_stove average	**98.91%**
Sample size	**92**
Total population size	**69,870**
Required precision	**95%**
z-value at 90% confidence	**1.96**
Confidence interval (+/-)	**2%**
Lower bound of the interval	**97%**
Higher bound of the interval	**101%**
Maximum error (precision)	**10%**
Sample monitoring precision	**2%**
Conclusion	**Precision OK**

2020 :

Ope_stove average	**98.89%**
Sample size	**90**
Total population size	**98,448**
Required precision	**95%**
z-value at 90% confidence	**1.96**
Confidence interval (+/-)	**2%**
Lower bound of the interval	**97%**
Higher bound of the interval	**101%**
Maximum error (precision)	**10%**
Sample monitoring precision	**2%**
Conclusion	**Precision OK**

Data / Parameter	μ_y

Data unit	Fraction
Description	Adjustment to account for any continued use of pre-project devices during year y
Value applied:	15.07 (average) Detail per batch : <table><tr><td>batch</td><td>µ</td></tr><tr><td>2018</td><td>13.38%</td></tr><tr><td>2019</td><td>17.02%</td></tr><tr><td>2020</td><td>14.65%</td></tr></table>
Comments	During the annual monitoring campaign, mandated field agents inquire if the baseline stove that was supposed to be replaced by the ICS is still used. Field agents estimate the usage rate of the pre-project stove(s) by formulating questions to determine the frequency of usage of both the project devices and baseline devices. Results per batch for µ parameter, and associated precision levels :

2018 batch

Batch average	13.38%
Sample size	90
Total population size	35,142
Required precision	95%
z-value at 90% confidence	1.96
Confidence interval (+/-)	7.0%
Lower bound of the interval (1-p)	6%
Higher bound of the interval (1-p)	20%
Maximum error (precision)	10%
Sample monitoring precision	8.11%
Conclusion	Precision OK

2019 batch

Batch average	17.02%
Sample size	92
Total population size	69,870
Required precision	95%
z-value at 90% confidence	0.95
Confidence interval (+/-)	3.7%
Lower bound of the interval (1-p)	13%
Higher bound of the interval (1-p)	21%
Maximum error (precision)	10%
Sample monitoring precision	4%

	Conclusion	Precision OK	

Batch average	14.65%	
Sample size	90	2020 batch
Total population size	98,448	
Required precision	95%	
z-value at 90% confidence	1.96	
Confidence interval (+/-)	7.3%	
Lower bound of the interval (1-p)	7%	
Higher bound of the interval (1-p)	22%	
Maximum error (precision)	10%	
Sample monitoring precision	9%	
Conclusion	Precision OK	

Data / Parameter	$\eta_{new,i,j}$
Data unit	%
Description	Efficiency of the device of each type i and batch j implemented as part of the project activity.
Value applied:	30.68 (CPA1 vintage-weighted efficiency average) Details per batch : batch / Monitored η_{new} / Final value 2018 / 31.04% / 29.5% 2019 / 30.98% / 29.5% 2020 / 30.34% / 29.5%
Comments	Tests were performed by third party KNUST technology consultancy center (University of Kumasi) with calibrated equipment and cross checked with manufacturer information. WBT sensors equipment (carbon Monoxide sensors, carbon dioxide sensors, duct temperature, water temperature) were calibrated in March 2022. The scale has been calibrated in February 2022 also by CSIR, with a certificate valid until February 2023 (hence covering the monitoring period). All the equipment (sensors, scale) were calibrated by Council for Scientific and Industrial Research (CSIR) and 95% precision has been met as per calibration certificates provided by the KNUST center.

Details per batch:

batch	Monitored η_{new}	Final value
2018	31.04%	29.5%
2019	30.98%	29.5%
2020	30.34%	29.5%

	Sampling test have been conducted following a 95/10 precision in accordance with the "Standard for sampling and surveys for CDM project activities and programme of activities", and 95/10 precision was met for all batches. As per registered CPA-DD, option c (*"Determine the rate of efficiency drop for a representative sample of the first batch of project device i in year y and assume that same rate of loss in efficiency applies to all other batches."), the efficiency of the batches should be assessed using the drop of efficiency of the first batch. However, results have shown that efficiency was not dropping by the year as expected. In reality the efficiency was increasing with the year as per table below:*

Batch	MP1	MP2	MP3	MP4
2018	29.97%	30.49%	33.92%	31.04%
2019	-	30.27%	32.77%	30.98%
2020	-	-	30.80%	30.34%

The older the stoves age, the more efficient they become – up to a certain level before efficiency drops because of breaks on the liner. This is due to the fact that ceramic stoves while becoming dryer thanks to usage becomes also more efficient. This was also observed under the VPA1/2/3 of Man&Man enterprise PoA under Gold standard. WBTs results of those VPAs are also added in the ER sheet.

It is therefore impossible to account for a dropdown in efficiency, while the efficiency is rising.

However, to be conservative and consistent with the CPA-DD. PP has opted to cap the efficiency for this monitoring period, at the efficiency tested by the KNUST technology consultancy center at CDM validation stage (29.5%). Therefore, 29.5% is the value used for this monitoring period.

Data / Parameter	Date of commissioning of batch j
Data unit	Date

Description	To establish the date of commissioning, the Project Participant opts to group the devices in "batches" and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch
Value applied:	2018 batch : from 04/06/2018 until 03/06/2019 2019 batch : from 04/06/2019 until 03/06/2020 2020 batch ; from 04/06/2020 until 03/06/2021
Comments	Excel spreadsheet provided to the VVB

Data / Parameter	**Date of commissioning of project device i**
Data unit	Date
Description	Actual date of commissioning of the project device.
Value applied:	Excel spreadsheet provided to the VVB
Comments	-

Data / Parameter	$N_{d,HH}$
Data unit	Number
Description	Number of project devices distributed per household
Value applied:	1.17 (all-surveys average)
Comments	Only one cooking stove per household is registered in the electronic database. If a household purchases more than one cooking stoves, monitoring surveys of sampled kitchens' stoves in use will account for any additional project device and be reflected in adjustment factor $N_{d,HH}$

Data / Parameter	N
Data unit	Number

Description	Number of project devices distributed
Source of data	Internal database
Description of measurement methods and procedures applied	Every time an ICS is sold a sale agreement is filled and an electronic database is filled. Based on the information collected into this electronic database, the number of ICSs distributed is determined.
Frequency of monitoring/recording	Recorded at the time of commissioning/distribution of project devices
Value applied:	203,460
Monitoring equipment	No equipement involved Every time an ICS is sold a sale agreement is filled and an electronic database is filled. Based on the information collected into this electronic database, the number of ICSs distributed is determined.
QA/QC procedures applied	--
Purpose of data	Calculation of Emission Reductions calculation
Calculation method	Sum of all project devices distributed.
Comments	--

7.2 Baseline Emissions

Emission reductions are calculated as:

$$ER_y = \Sigma_i \Sigma_j ER_{y,i,j} - LE_y$$

Where:

i = Indices for the situation where more than one type of project device is introduced to replace the pre project devices devices[11]

j Indices for the situation where there is more than one batch of project device

ER_y Emission reductions during year y in t CO_2e

[11] *For example, in some instances, full replacement of the pre-project device would require the implementation of more than one project device (e.g. one stove suitable for cooking and the other stove suitable for cooking/boiling water).*

$ER_{y,i,j}$ Emission reductions by project device of type i and batch j during year y in t CO_2e

LE_y Leakage emissions in the year y

$$ER_{y,i,j} = B_{y,savings,i,j} \times N_{y,i,j} \times \mu_y \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected\_fossil\ fuel}$$

Where:

$B_{y,savings,i,j}$ Quantity of woody biomass that is saved in tonnes per cook stove of type I and batch j during year y

fNRB,y Fraction of woody biomass saved by the project activity in year y (established as non-renewable biomass using survey methods)

$NCV_{biomass}$ Net calorific value of the non-renewable woody biomass that is substituted (IPCC default for wood fuel, 0.015 TJ/tonne, based on the gross weight of the wood that is 'air-dried')

$EF_{projected\_fossilfuel}$ Emission factor for the fossil fuels projected to be used for substitution of non-renewable woody biomass by similar consumers. Use a value of 81.6 tCO_2/TJ

$N_{y,j,j}$ Number of project devices of type i and batch j operating during year y

μ_y Adjustment to account for any continued use of pre-project devices during the year y applying equation 6

$$B_{y,savings,i,j} = B_{old,i,j} \times (1 - \eta_{old,i,j} / \eta_{new,i,j})$$

Where:

$\eta_{old,i,j}$ Efficiency of the old devices being replaced by project devices of type i and batch j.

$\eta_{new,i,j}$ Efficiency of the project device i and batch j

$B_{old,i,j}$ is determined as follows:

$$B_{old,i,j} = B_{old,HH} \div N_{d,HH}$$

Where:

$B_{old,HH}$ = Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices (tonnes/household/year)

$N_{d,HH}$ = Number of project devices per household (number)

The formula above accounts the monitored number of project devices per household in the calculation (detailed and complete calculation available on ER sheet).

Bold,i,j is multiplied by a net to gross adjustment factor of 0.95 to account for leakages, in which case surveys are not required.

Jiko Matawi stove stove: Substituting for values for Jiko Marawi stoves in use for 100% of the time over the monitoring period, the calculation results in the below.

Batch 2018 :

	Value	Unit	Source/reference
N	35,142	n/a	Internal database
$N_{y,i,j}$	32,018	n/a	Calculated $N_{y,i,j}$ = N * pop_stoves,y
$\mu_{,y}$	0.1338	fraction	Monitoring surveys (see section 6.1 above)
$f_{NRB,y}$	0.9884	fraction	Section B.4.1 of registered CPA1
$NCV_{biomass}$	0.015	TJ/tonne	Section B.4.2 of registered CPA1
$EF_{projected\_fossilfuel}$	81.6	tCO_2/TJ	IPCC Guidelines 2006
$B_{y,savings,i,j}$	1.40	tonnes/year	(detail per batch on the ER sheet) $B_{y,savings,i,j} = B_{old,i,j}*(1 - \eta_{old,i,j}/\eta_{new,i,j})*0.95$
$B_{old,i,j}$	3.50	tonnes/year	Section B.4.2 of registered CPA1
$B_{old,HH}$	4.32	tonnes/HH/year	Section B.4.2 of registered CPA1
$N_{d,HH}$	1.23	devices/HH	Monitoring surveys (see section 6.1 above)
$\eta_{old,i,j}$	18	%	Section B.4.2 of registered CPA1
$\eta_{new,i,j}$	29.5	%	See section 7.1 above and ER sheet
Baseline Emissions	61,527	tCO_2/year	Section F.1, ER calculations $ER_{y,i,j} = B_{y,savings,i,j} \times N_{y,i,j} \times \mu_y \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected\_fossil\ fuel}$
Project emissions (PE_y)	0	tCO_2/year	See section 6.3 below
Leakage emissions	0	tCO_2/year	See section 6.4 below

Batch 2019 :

	Value	Unit	Source/reference
N	69,870	n/a	Internal database
$N_{y,i,j}$	69,111	n/a	Calculated $N_{y,i,j}$ = N * pop_stoves,y
$\mu_{,y}$	0.1702	fraction	Monitoring surveys (see section 6.1 above)
$f_{NRB,y}$	0.9884	fraction	Section B.4.1 of registered CPA1
$NCV_{biomass}$	0.015	TJ/tonne	Section B.4.2 of registered CPA1
$EF_{projected\_fossilfuel}$	81.6	tCO_2/TJ	IPCC Guidelines 2006
$B_{y,savings,i,j}$	1.45	tonnes/year	(detail per batch on the ER sheet) $B_{y,savings,i,j} = B_{old,i,j}*(1 - \eta_{old,i,j}/\eta_{new,i,j})*0.95$

$B_{old,i,j}$	3.65	tonnes/year	Section B.4.2 of registered CPA1
$B_{old,HH}$	4.32	tonnes/HH/year	Section B.4.2 of registered CPA1
$N_{d,HH}$	1.18	devices/HH	Monitoring surveys (see section 6.1 above)
$\eta_{old,i,j}$	18	%	Section B.4.2 of registered CPA1
$\eta_{new,i,j}$	29.5	%	See section 7.1 above and ER sheet
Baseline Emissions	132,448	tCO_2/year	Section F.1, ER calculations $ER_{y,i,j} = B_{y,savings,i,j} \times N_{y,i,j} \times \mu_y \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected\_fossil\ fuel}$
Project emissions (PE_y)	0	tCO_2/year	See section 7.3 below
Leakage emissions	0	tCO_2/year	See section 7.4 below

Batch 2020 :

	Value	Unit	Source/reference
N	94,016	n/a	Internal database
Ny,i,j	92,969	n/a	Calculated $N_{y,i,j}$ = N * pop_stoves,y
μ,y	0.1465	fraction	Monitoring surveys (see section 6.1 above)
$f_{NRB,y}$	0.9884	fraction	Section B.4.1 of registered CPA1
$NCV_{biomass}$	0.015	TJ/tonne	Section B.4.2 of registered CPA1
$EF_{projected\_fossilfuel}$	81.6	tCO_2/TJ	IPCC Guidelines 2006
$B_{y,savings,i,j}$	1.60	tonnes/year	(detail per batch on the ER sheet) $B_{y,savings,i,j} = B_{old,i,j} * (1 - \eta_{old,i,j}/\eta_{new,i,j}) * 0.95$
$B_{old,i,j}$	4.14	tonnes/year	Section B.4.2 of registered CPA1
$B_{old,HH}$	4.32	tonnes/HH/year	Section B.4.2 of registered CPA1
$N_{d,HH}$	1.04	devices/HH	Monitoring surveys (see section 6.1 above)
$\eta_{old,i,j}$	18	%	Section B.4.2 of registered CPA1
$\eta_{new,i,j}$	29.5	%	See section 7.1 above and ER sheet
Baseline Emissions	208,082	tCO_2/year	Section F.1, ER calculations $ER_{y,i,j} = B_{y,savings,i,j} \times N_{y,i,j} \times \mu_y \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected\_fossil\ fuel}$
Project emissions (PE_y)	0	tCO_2/year	See section 7.3 below
Leakage emissions	0	tCO_2/year	See section 7.4 below

7.3 Project Emissions

As per ER calculation, and the methodology, there are no project emissions considered.

7.4 Leakage

As per AMS II.G, leakage is already taken account of in the calculation of baseline emissions by multiplying $B_{y,savings,i,j}$ by a net to gross adjustment factor of 0.95 to account for leakages. In such case, surveys to determine leakage ex-post are not required.

7.5 Net GHG Emission Reductions and Removals

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
2021 (01-January-2021 – 31-December-2021)	284,401	0	0	284,401
2022 (01-January-2022 – 31-May-2022)	117,656	0	0	117,656
Total	**402,057**	0	0	**402,057**

The monitoring period for which GHG emission reductions were achieved spans from 01-January-2021 to 31-May 2022.

Ex-ante emissions reductions/removals	Achieved emissions reductions/removals	Percent difference	Justification for the difference
355,212[12] Ratio ERs/stove/year : 1.23	402,057 Ratio ERs/stove/year : 1.40	11.7%[13]	Although some parameters were slightly overestimated in the ex-ante calculation like $B_{old,i,j}$. The overall emissions reductions are higher than what was estimated Ex-ante. This is mainly due to monitored parameters like μ y, $\eta_{new,i,j}$ and $p_{operational\ stoves}$, which were found higher than expected. These parameters were initially estimated lower for conservativeness reasons. These higher values of monitored parameters caused a higher ratio of ERs/stove/year hence a higher total volume of ERs.

[12] Ex-ante emissions reductions/removals for this monitoring period were calculated based on the ex-ante ER calculation values of 2021 to which were added the 2022 values multiplied by a 151/365 factor, counting for the proportion of the year covered by the monitoring period.

[13] Percent difference is calculated as following : % difference = (achieved emissions reductions – ex-ante emissions reductions) / achieved emission reductions * 100